

COGNITIVE COMPLIANCE & FORENSIC AUDITOR

Multi-Agent Courtroom — Audit Report

Transaction ID	TXN-2026-XB-004821	Prediction	HIGH RISK
Amount	\$47,500.00 (INR → USD)	Confidence	94%
Timestamp	2026-03-15T08:42:11Z	Frozen	True
Run ID	6059267f-035e-4e25-bc05-aa77493d6071	SARs Filed	2
Report Generated	2026-03-16T13:44:34.953360+00:00		

1. AUDITOR FINAL VERDICT

AUDITOR FINAL RULING: TRANSACTION FROZEN

Both jurisdictional experts are in agreement: this transaction is HIGH RISK and NON-COMPLIANT across RBI (India) and EU/AMLD-6 frameworks.

KEY AGGRAVATING FACTORS:

- GNN cycle motif CYCLE-A detected — textbook smurfing pattern.
- Shell company probability 87% (above both RBI and EU thresholds).
- Sanctions hit on intermediary ACC-AE-0077 (UAE).
- Both end-point accounts have PENDING KYC and unverified beneficial owners.
- 94-day and 61-day old accounts — both below the 180-day safe-harbour.

ACTIONS ORDERED:

- Immediately freeze TXN-2026-XB-004821.
- File STR with FIU-IND (India) within 24 hours.
- File SAR with GoAML (EU) within 24 hours.
- Escalate ACC-AE-0077 to the Sanctions Compliance team.
- Place a 90-day monitoring hold on cluster CLUSTER-SHELL-07.

CONFIDENCE: 94%

XAI REPORT: Attached.

2. JURISDICTIONAL EXPERT VERDICTS

India Expert (RBI)

VERDICT: HIGH RISK – NON-COMPLIANT

Under RBI jurisdiction I find three critical breaches:

- [RBI-KYC-2016-§38] Amount \$47,500 exceeds ₹25 lakh EDD threshold. No source-of-funds declaration is on file. BREACH.
- [RBI-FEMA-2013-§6(3)] Two intermediary hops (Singapore, UAE) within 72 hours trigger mandatory FIU-IND reporting. BREACH.
- [RBI-AML-2023-§12] Sender account age = 94 days. Transaction exceeds ₹15 lakh monthly cap for unverified new accounts. BREACH.

Sender KYC status: PENDING. Beneficial owner: UNVERIFIED.

RECOMMENDATION: Freeze transaction. File STR with FIU-IND within 24 hours.

EU Expert (AMLD-6 / AI Act)

VERDICT: HIGH RISK – PARTIAL BREACH

Under EU/AMLD-6 and EU AI Act jurisdiction:

1. [AMLD6-Art3(4)] GNN shell-company probability = 87%, exceeding the 80% threshold. A SAR must be filed with GoAML within 24 hours. BREACH.
2. [EUAIA-2024-Art22] The CCFA system is generating this XAI report, satisfying the explainability obligation. COMPLIANT.

Intermediary account ACC-AE-0077 (UAE) has a SANCTIONS HIT and PEP flag in KYC database — this is an aggravating factor under AMLD-6 Art 18.

RECOMMENDATION: Freeze transaction. Mandatory SAR via GoAML. Escalate ACC-AE-0077 to sanctions compliance team.

3. REGULATORY BREACH SUMMARY

Rule ID	Status	Reason	Penalty Cap
RBI-KYC-2016-§38	BREACH ✗	Amount ≈\$47,500 (>\$30k threshold). No source-of-funds document updated.	Up to 1% of global annual turnover or ₹1.5 billion.
RBI-FEMA-2013-§6(3)	BREACH ✗	2 intermediate hops (Singapore, UAE) within 72 hrs.	Up to 4% of global annual turnover or ₹1.5 billion.
RBI-AML-2023-§12	BREACH ✗	Sender account age = 94 days; transaction amount exceeds cap.	Up to 4% of global annual turnover or ₹1.5 billion.
EUAIA-2024-Art22	COMPLIANT ✓	CCFA system generates XAI report per this rule — compliant.	Up to €10 million or 2% of worldwide annual turnover.
AMLD6-Art3(4)	BREACH ✗	Shell company probability = 87%. SAR not yet filed.	Up to €10 million or 2% of worldwide annual turnover.

4. SHAP FEATURE ATTRIBUTIONS (XAI)

Base value (population average risk): 0.350 | Final model output: 0.940 | Model: Pytorch Geometric GAT v2.1.0-mock

Feature	SHAP Value	LIME Weight	Direction	Raw Value
Circular money flow (smurfing cycle) detected	+0.4230	0.4441	▲ RISK	[{'cycle_id': 'CYCLE-2026-03-16T13:44:34.947262'}]
Overall graph anomaly score from GNN	+0.3458	0.3631	▲ RISK	0.91
Probability that sender/receiver is a shell company	+0.2784	0.2923	▲ RISK	0.87
Number of known bad actors in same cluster	+0.2324	0.2440	▲ RISK	3
Transaction velocity spike (3× 48-hr baseline)	+0.2125	0.2231	▲ RISK	True
Suspicious intermediary routing hops	+0.1738	0.1825	▲ RISK	[{'account_id': 'ACC-2026-03-16T13:44:34.947262'}]
Amounts vary slightly to evade filters (smurfing)	+0.1520	0.1596	▲ RISK	True
Sender account is very new (< 180 days)	+0.1296	0.1361	▲ RISK	True
Receiver account is very new (< 180 days)	+0.1088	0.1142	▲ RISK	True
Device fingerprint matches known good device	-0.0610	0.0640	▼ SAFE	False
Sender provided a stated purpose for transfer	-0.0500	0.0525	▼ SAFE	False
Sender bank is a recognised Tier-1 institution	-0.0300	0.0315	▼ SAFE	False

5. LIME LOCAL EXPLANATION

LIME local approximation (neighbourhood size=500 perturbations):

The GNN classifier assigned HIGH RISK to transaction TXN-2026-XB-004821 primarily because similar transactions in its neighbourhood (±15% amount, ±30 days) with cycle motifs AND sender account age <120 days were classified HIGH RISK in 98.2% of cases. The most influential boundary feature is the presence of a confirmed cycle motif (CYCLE-A): removing it reduces the risk score from 0.91 to 0.43. The velocity spike (3× 48h baseline) adds an independent +0.18 to the risk logit. Beneficiary account age <90 days adds +0.14. These three features together account for 79% of the decision.

6. SAR / STR FILING RECEIPTS

Jurisdiction	Tracking ID	Filed At	Review Days
RBI	STR-FIUIND-20260316-99124	2026-03-16T13:44:34.947262+00:00	3
EU	SAR-GOAML-20260316-63358	2026-03-16T13:44:34.947262+00:00	3

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